# Algebra Monsoon 2026 Lecture 16 The Snake Lemma

## The Snake Lemma

**Def** A group $G$ is said to be **simple** if $G \neq \{e\}$ and if the only **normal** subgroups of $G$ are $\{e\}$ and $G$.

Analogy: A natural number $n$ is **prime** if $n \neq 1$ and the only divisors of $n$ are $1$ and $n$.

Example Every finite group of **prime order** is simple. (Converse not true)

**Proposition** (Jordan-Hölder) For every simple group $S$, there is a unique mapping $m_S : (\text{finite groups}) \longrightarrow \mathbb{N}$ such that (the multiplicity of $S$ in $G$)

1) $m_S(G) = 1 \iff G$ is isomorphic to $S$

2) $m_S(G) = m_S(G') + m_S(G'')$

whenever there is an exact sequence

$$\{e\} \longrightarrow G' \longrightarrow G \longrightarrow G'' \longrightarrow \{e\}$$

Analogy: For every prime $p$, there is a unique mapping $v_p : \mathbb{N}^* \longrightarrow \mathbb{N}$ such that

1) $v_p(n) = 1 \iff n = p$

2) $v_p(n) = v_p(n') + v_p(n'')$ whenever $n = n'n''$.

Def A pair $(f, g)$ of morphisms

$$G' \xrightarrow{f} G \xrightarrow{g} G''$$

of groups is said to be exact if $f_*(G') = g^*(\{e\})$.

$f_*(G')$: image of $f$; $g^*(\{e\})$: kernel of $g$

In particular, the subgroup $f_*(G')$ is normal in $G$.

A sequence of morphisms

$$\cdots \xrightarrow{f_{n-2}} G_{n-1} \xrightarrow{f_{n-1}} G_n \xrightarrow{f_n} G_{n+1} \xrightarrow{f_{n+1}} G_{n+2} \xrightarrow{f_{n+2}} \cdots$$

is said to be exact if $(f_n, f_{n+1})$

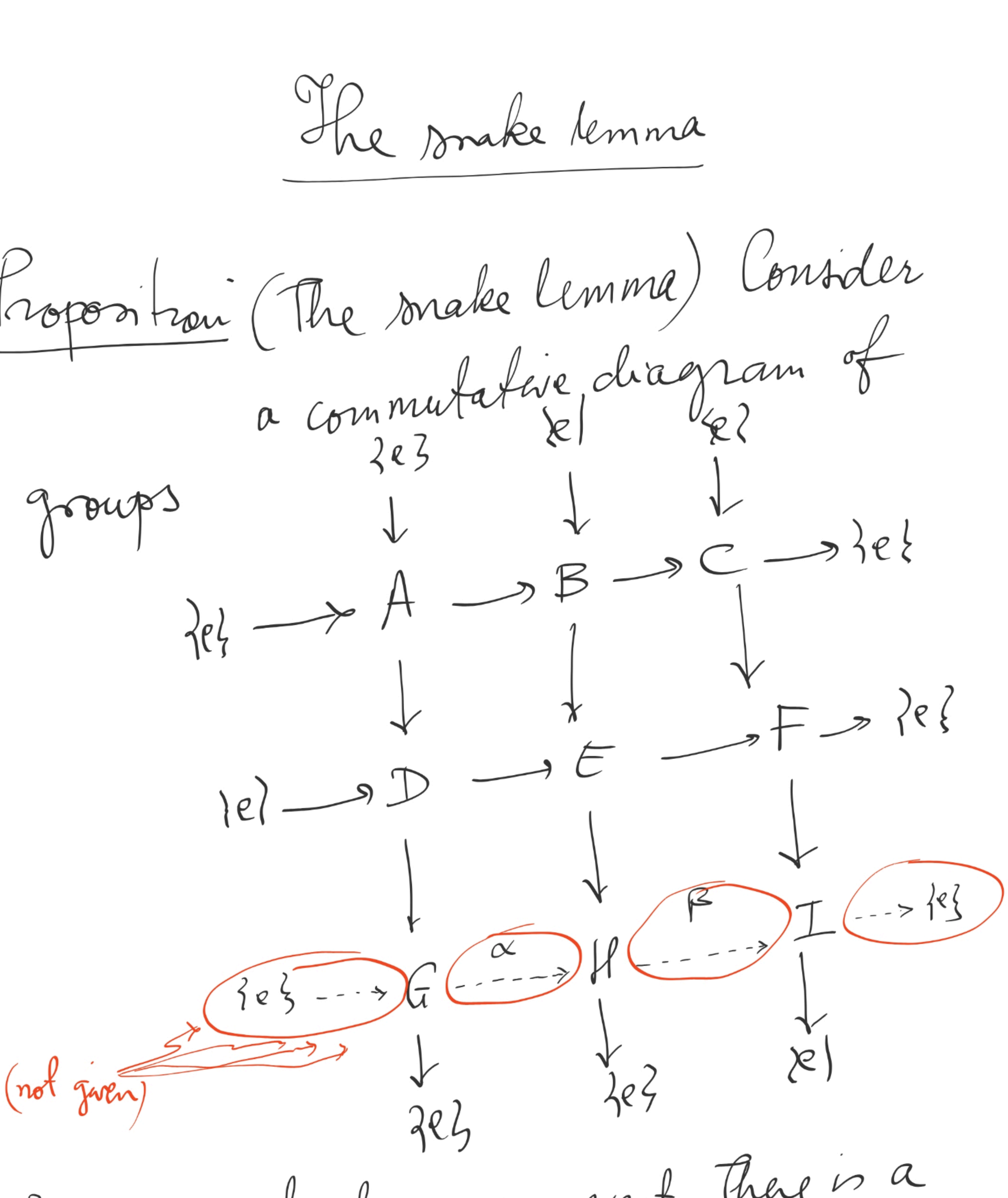
is exact for every $f$?

Ex 1) $\{e\} \to H \xrightarrow{f} G$ is exact
$\iff f$ is injective

2) $G \xrightarrow{g} H \to \{e\}$ is exact
$\iff g$ is surjective.

# The snake lemma

**Proposition** (The snake lemma) Consider a commutative diagram of groups




Assume the bottom two rows are exact. There is a

whose rows and columns are exact. There is a unique morphism of groups $\alpha : G \to H$ making the square

$$\begin{array}{ccc} D & \to & E \\ \downarrow & & \downarrow \\ G & \xrightarrow{\alpha} & H \end{array}$$

commute, and a unique morphism $\beta : H \to I$ making the square

$$\begin{array}{ccc} E & \to & F \\ \downarrow & & \downarrow \\ H & \xrightarrow{\beta} & I \end{array}$$

commute. Moreover, the sequence

$$\{e\} \to G \xrightarrow{\alpha} H \xrightarrow{\beta} I \to \{e\}$$

is exact.

To define $\alpha$, let $g \in G$. As $D \to G$ is surjective, there exists $d \in D$ such that $d \mapsto g$. Let $h$ be the image of $d$ under $D \to E \to H$. Let us check that $h$ depends only on $g$, not on the choice of $d$. Suppose that $d' \in D$ is another element such that $d' \mapsto g$ in $G$ and $d' \mapsto h'$ under $D \to E \to H$. Now, $d'd^{-1}$ is in the kernel of $D \to G$ so there exists an $a \in A$ such that $a \mapsto d'd^{-1}$ in $D$. Consequently, $a$ and $d'd^{-1}$ have the same image in $H$. The image of

have the $B \to E \to H$ exact, $\alpha$ is trivial(e) because $B \to E \to H$ is exact, and the image of $d'd^{-1}$ is $h'h^{-1}$. Hence $h = h'$. We may thus define $\alpha(g) \in H$, for $g \in G$, to be the image of any $d \in D$ such that $d \mapsto g$. The mapping $\alpha : G \to H$ exists because of the exactness of the first two rows, and it is the only mapping making the square

$$\begin{array}{ccc} D & \longrightarrow & E \\ \downarrow & & \downarrow \\ G & \xrightarrow{\alpha} & H \end{array}$$

commute.

Let us show that $\alpha$ is a morphism of groups: $\alpha(g_1 g_2) = \alpha(g_1)\alpha(g_2)$ for $g_1, g_2 \in G$. Choose $d_1, d_2 \in D$ such that $d_1 \mapsto g_1$ and $d_2 \mapsto g_2$. For $g_1 g_2$, we may choose $d_1 d_2$, because $D \to G$ is a morphism of groups. Now, $\alpha(g_1)$ is the image of $d_1$, $\alpha(g_2)$ is the image of $d_2$, and $d_1 d_2$ is the image under the composite morphism $D \to E \to H$

$\alpha(g_1 g_2)$ ——— $\alpha_1 \alpha_2 / B \Rightarrow$

But this composite is a morphism of groups, hence $\alpha(g_1 g_2) = \alpha(g_1)\alpha(g_2)$

We have proved the existence and uniqueness of $\alpha$ using the exactness of the first two columns.

The same argument, using this time the exactness of the last two columns, gives the existence and uniqueness of $\beta$.

It remains to show that the sequence

(1) $\alpha^*(\{e\}) = \{e\}$

(2) $\alpha_*(G) = \beta^*(\{e\})$,

and (3) $\beta_*(H) = I$.

(1) • Suppose $\alpha(g) = 1$ for some $g \in G$.

$g \mapsto 1$

• Choose $b \in B$ such that $\begin{matrix} b \\ \downarrow \\ g \end{matrix}$.

Let $b \mapsto e$.

$\begin{matrix} e \\ \downarrow \end{matrix}$, so there exists $b \in B$ such that $\begin{matrix} b \\ \downarrow \\ e \end{matrix}$

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- Let $b \mapsto c$. Then $\begin{array}{c} C \\ \downarrow \\ 1 \end{array}$ became

$$\begin{array}{ccc} b & \mapsto & c \\ \downarrow & & \downarrow \\ d & \mapsto & e \mapsto 1 \end{array}$$

Therefore $c = 1$.

- Therefore there is an $a \in A$ such that $a \mapsto b$.

Let $\begin{array}{c} a \\ \downarrow \\ d' \end{array}$

- As $d' \mapsto e$, we have $d = d'$.

- Hence $g = 1$, since $\begin{array}{c} a \\ \downarrow \\ d \\ \downarrow \\ 1 \end{array}$.

(2) $\alpha_*(G) \overset{?}{=} \beta^*(1)$.

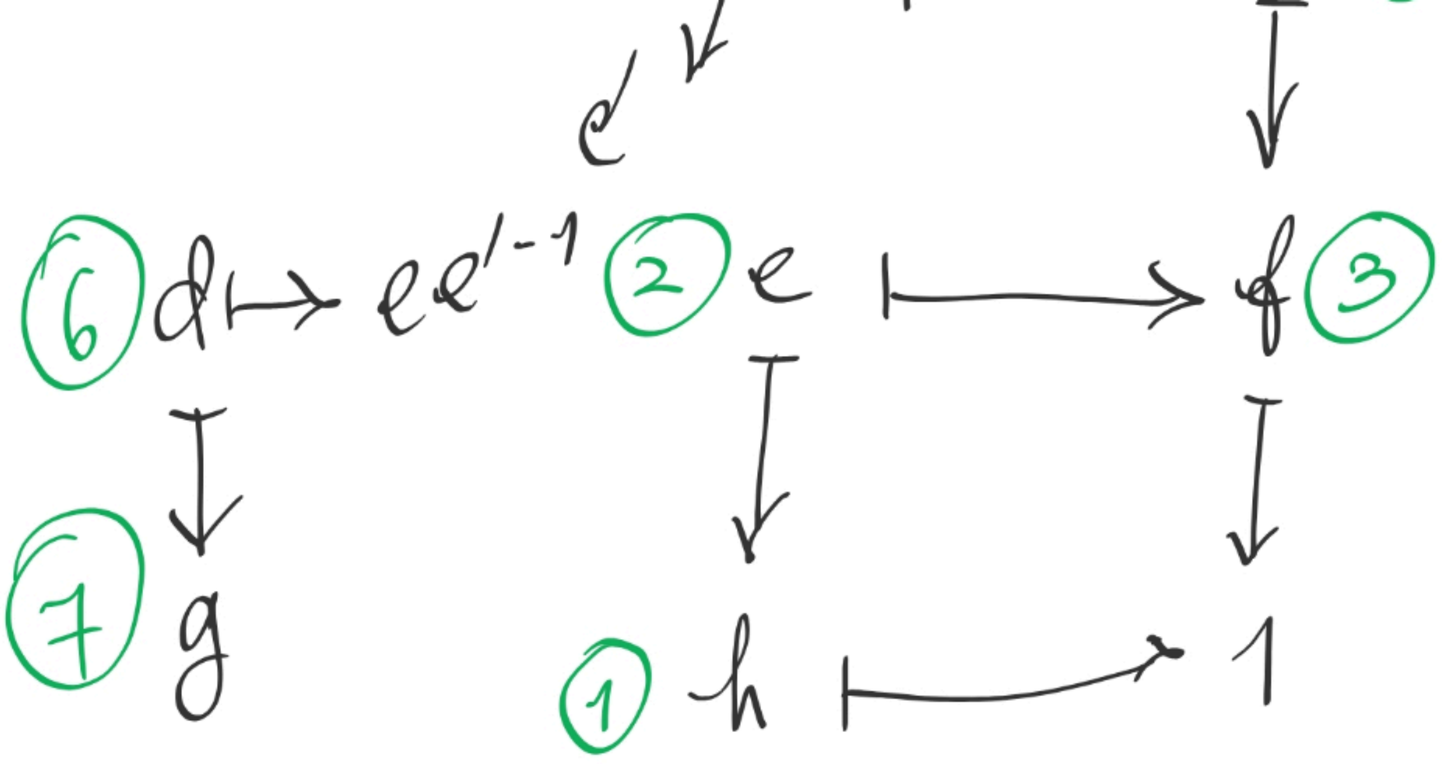
We have $\alpha_*(G) \subset \beta^*(1)$

because

$$\begin{array}{ccccc} (2)\ d & \longmapsto & (3)\ e & \longmapsto & 1\ (4) \\ \downarrow & & \downarrow & & \downarrow \\ (1)\ g & \longmapsto & \alpha(g) & \longmapsto & \beta(\alpha(g)) \end{array}.$$

We have $\beta^*(1) \subset \alpha_*(G)$ because

$$\begin{array}{ccccc} (5)\ b & \longmapsto & c\ (4) & & \\ \downarrow & & \downarrow & & \\ a & \longmapsto & \alpha(a) & \longmapsto & \beta(\alpha(a)) \end{array}$$




There exists $d \in D$ such that $d \mapsto e e'^{-1}$.

Let $d \downarrow g$. Then $d \mapsto e e'^{-1}$, $g \mapsto h$.

so $\alpha(g) = h$. Therefore $\alpha_*(G) = \beta^*(1)$.

Finally, $\beta$ is surjective because $E \to F$ and $F \to I$ are surjective and

$$\begin{array}{ccc} E & \longrightarrow & F \\ \downarrow & & \downarrow \\ H & \xrightarrow{\beta} & I \end{array}$$

commutes. □